



Buy a gaming laptop

This guide tells you everything you need to know to make a smart decision when buying a gaming laptop. <http://dgit.in/sep20-41>



Neanderthals and climate

Sudden climate change shortly before the disappearance of Neanderthals made them create better tools. <http://dgit.in/sep20-42>

THE CANDIDATES

We have a mixed bag of boards in this mini-comparison. There are three premium boards from the big three manufacturers, however, these aren't the absolute best, rather, they can be termed the penultimate best as they all have a board or two above them in the SKU stack.



ASUS ROG STRIX B550-F GAMING + WI-FI

ASUS's boards have always been a little better looking given

their balanced aesthetics. The B550-F Wi-Fi is one of the more expensive boards you can get with the chipset and they do make it worth the money. For starters, this is the only board we've received which has an AX200 CNVi Wi-Fi module. The rest are all Wi-Fi 5 or 802.11ac, if they do have Wi-Fi. For the LAN port it has a 2.5GbE Intel I225-V NIC which is better than the Realtek solutions we tend to see on the other boards.

The board uses an 8-Pin and 4-Pin EPS connector for the CPU power which feeds into an ASUS ASP1106 VRM controller. It implements a 12+2 phase design even though the ASP1106 is only capable of 4+2 mode. And there are no doublers in place so the board is splitting the VRM across multiple MOSFETs and triggering them in unison. Lastly, it uses Vishay SiC639 50A for the power stages.



GIGABYTE B550 AORUS PRO

The GIGABYTE B550 AORUS Pro that we received is the one without Wi-Fi. GIGA-

BYTE uses a fin stack for one half of the VRM heatsink which is much better appreciated than solid metal block heatsinks owing to the larger surface area. This is the only board with three full-length PCIe x16 slots, although only the first one is wired for x16 lanes whereas the other two are wired for x4 and x2 lanes. For the networking part,

GIGABYTE uses the Realtek 8125BG 2.5 GbE NIC. Another great thing about the GIGABYTE B550 Aorus Pro is that it has 12 USB ports and of different varieties on the rear I/O while most boards tend to max out at 8.

Coming to the power circuitry, we see an 8-Pin EPS connector driving a Renesas RAA229004 PWM controller. It's paired with Intersil ISL6617A doublers to form a 12+2 arrangement and uses Vishay SiC651C 50A power stages. Like the ASUS board, the B550 AORUS Pro can also boast of a robust VRM.



MSI MAG B550 TOMAHAWK

The Tomahawk series from MSI are really popular across different platforms. With

the B550 chipset, MSI is hoping to achieve the same success that it has had with the other boards in the Tomahawk line. MSI opts for dual LAN ports on this board using Realtek RTL8111H 1 GbE and RTL8125B 2.5 GbE NICs. They can also be teamed. The board has two full-length PCIe x16 slots with one wired for x16 lanes and the other for x8 lanes. It also has fewer USB ports than the other boards in its segment.

For power delivery, the MSI MAG B550 Tomahawk uses a Renesas RAA229004 which is configured in a 5+2+1 mode and does not use a doubler. Instead it has 2x Intersil ISL99360 60A per phase and they're individually controlled. The effective VRM arrangement becomes 10+2+1 which is a little odd but works just as well as the other boards that we've seen.



MSI B550M PRO-VDH WIFI

We had this one lying around for a different project and we threw it into the mix just for fun. Being a more budget-oriented board, this board is very likely to be extremely popular. It's a Micro-ATX board with just one PCIe x16 slot and two x1

slots. However, it has more display connectors, the same set of M.2 ports as the rest of the boards but fewer USB ports. Even the audio section is a downgrade in terms of the CODEC as well as the number of audio jacks available. It compensates for its shortcomings by adding dual-band Wi-Fi 5 and Bluetooth 4.2.

The B550M PRO-VDH Wi-Fi has a simple VRM circuitry, it uses a Richtek RT8894 configured in a 4+2 mode and uses On Semi 4C029N in combination with the On Semi 4C024N. Just enough to get the job done. The rudimentary VRM is quite cool even when a high-end Ryzen 9 processor is plugged in, so that's saying something.

VERDICT

While we didn't see much of a performance difference with the different B550 boards when paired with a Ryzen 9 3900XT, we did get to see how well the VRMs turned out to be. The ASUS ROG Strix B550-F Gaming + Wi-Fi was by far the coldest board followed by the MSI MAG B550 Tomahawk and then came in the GIGABYTE B550 Aorus Pro. The MSI and GIGABYTE boards were very close in load temperatures with the exception of the PRO-VDH Wi-Fi. From a features perspective, the ASUS and GIGABYTE boards are great offerings. We really prefer the ASUS board for the fact that it includes a Wi-Fi 6 CNVi card, slightly better CODEC and a great VRM. The GIGABYTE board, on the other hand, has a ridiculous amount of USB ports, three full-length x16 slots and has a lot of headers for cooling fans and features metal brackets over the RAM and PCIe slots. It's also the cheapest of the three premium boards. So if you're looking for a little future-proofing, then the ASUS ROG Strix B550-F Gaming + Wi-Fi board is our recommendation whereas if you want a great feature set for a good price, the GIGABYTE B550 Aorus Pro and MSI MAG B550 Tomahawk are good picks. If you're broke like the rest of us, there's always the MSI B550M PRO-VDH Wi-Fi for the best value. 